Network Swiss Army Knife (NSAK)

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The Challenge

Rising Cyber Threats

- Attackers already leverage AI — defenders must keep pace
- Red & blue team exercises demand deep expertise
- High recurring effort — inaccessible for small and medium-sized enterprises
- Scripted automation is brittle and inflexible

Hypothesis

Research Questions

- Can an AI agent match a scripted scenario at automated network reconnaissance?
- How effectively can AI assist a human operator during a live security investigation?
- Does it hold up in real-world networks beyond a controlled lab?

Our Approach

Agentic AI

- Building upon the predecessor project: Network Swiss Army Knife
  NSAK — A modular framework for network security scenario automation
- Integrate an LLM agent into NSAK with LangChain
- Equip it with tool calling for autonomous execution
- Benchmark against a scripted reference scenario

3 LLMs • 2 Network Environments • Static Reference Baseline

What We Built

- AI Reconnaissance Scenario
- Static Reference Scenario
- Reproducible Network Environment
- Benchmark Suite
- And more...

Evaluation Highlights

Containerlab (Virtual)

- ✔ All 3 models completed reconnaissance autonomously
- ✔ Multi-agent decomposition improved smaller models significantly
- – Larger models showed no benefit from decomposition

BFH Network Security Lab (Physical)

- ✔ Frontier model navigated a live network with 35+ hosts
- – Structured output failed — adapted to unstructured mode
- – Small local model (qwen3:8b) could not complete the task

Static scenario: fast & reproducible, but no reasoning or report generation *“AI and scripted automation are complementary — not competing — approaches.”*

Demo: Interactive AI Agent

Approval workflow

- Operator confirms each drill before execution

AI reasoning

- Model selects tools, adapts strategy, explains decisions

Email report

- Prioritized findings sent as a human-readable summary

Demo: Interactive AI Agent

Prompt:

Help met to find all vulnerable services on interface eth1

Demo: Interactive AI Agent

Prompt:

Please check if the SMB share is insecure or if you can access user accounts from LDAP with weak credentials

Demo: Interactive AI Agent

Prompt:

I think we gathered enough information now, create a summary in Markdown and send it as an email

Key Findings

AI Scenario

- ✓ Prioritized, natural-language reports
- ✓ Adaptive to unknown environments
- ✓ Low barrier — minimal expertise needed
- – Variable output, hallucination risk
- – Requires human supervision

Static Scenario

- ✓ Scripted & reproducible
- ✓ Fast, no token cost
- ✓ Precise, auditable steps
- – No reasoning or report generation
- – Requires domain expertise to build

“AI and scripted automation are complementary — not competing — approaches.”